Design of Enterprise IoT

Tutorial 2 (Selecting IoT Devices)

Read the following use case and choose essential IoT devices to be used in the proposed solution.

IoT Use Case: Intelligent Port System for Fraud Detection

Port operations play a crucial role in global trade, handling vast amounts of cargo daily. However, fraudulent activities, such as cargo theft, pose significant financial and operational risks. According to China Police Daily, between 2013 and 2017, Chinese Public Security Traffic Management Authorities cracked over 4,000 criminal cases related to port logistics (Song et al., 2020). Cargo theft is often considered a "silent crime" because losses frequently go unreported and may only be discovered weeks or even months later.

One common cargo theft tactic involves manipulating a truck's empty weight before entering the port. Fraudsters add materials like liquid tanks, cement, or scrap to artificially increase the truck's weight during the initial weigh-in. After getting the empty truck weighed, these materials are secretly removed, creating a weight discrepancy that allows the truck to load excess cargo undetected. Such fraudulent practices lead to financial losses, regulatory violations, and operational inefficiencies.

Traditional fraud detection methods rely heavily on manual inspections, post-event investigations, and rule-based monitoring systems, which struggle to detect anomalies in real time. These limitations highlight the need for an intelligent port system that enhances security, improves efficiency, and minimizes fraudulent activities. IoT technology offers a promising solution by enabling real-time tracking, automated data collection, and intelligent analytics, allowing ports to proactively detect fraud and optimise operations.

Proposed Solution: An IoT-based intelligent port system leverages advanced IoT technology and data analytics to enhance security and efficiency. Key components include:

- **Smart Weighbridge Systems:** Infrared sensors and weight sensors detect improper truck positioning and compare real-time weight with historical records, ensuring accuracy in cargo measurement.
- **Environmental Context Integration:** Weather sensors provide real-time rainfall data to detect fraudulent weight manipulation involving hidden water tanks.
- **GPS-Based Route Monitoring:** GPS tracking ensures trucks follow designated routes, detecting unauthorised stops or route deviations that may indicate fraud.
- **IoT-Enabled Cargo Surveillance:** Smart cameras, RFID scanners, and environmental sensors verify cargo handling, preventing unauthorised cargo movement.

Design Enterprise IoT

Activity: *Please refer to the "**IoT Device Profile template**" on Canvas and provide a list of essential IoT devices that can be used in the above proposed solution.*

[Note] You are asked to select off-the-shelf IoT devices.

You can work on your document by answering the following questions:

- Device name:
  - What is the name of each IoT device?
- Device type:
  - What is the type of each IoT device? (E.g., sensor, actuator, or a device of both)
- Feature of Interest
  - What "thing" is being observed and/or acted upon?
- Property
  - Which aspect(s) of the "thing" is/are being observed and/or acted upon?
- Characteristics (examples included but not limited to)
  - Measurement/actuation range
  - Frequency
  - Accuracy
  - Deployment
  - Battery lifetime
- Components
  - What are the components embedded within the device?
- Justification
  - Why is the device selected for the proposed IoT solution?

References

Song, R., Huang, L., Cui, W., Óskarsdóttir, M., & Vanthienen, J. (2020). Fraud Detection of Bulk Cargo Theft in Port Using Bayesian Network Models. *Applied Sciences*, *10*(3). https://doi.org/10.3390/app10031056